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Question

Consider the vectors $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 0 \\ 3\sqrt{2} \end{pmatrix}$. For real-valued scalar variable x , the value of $\min_x \|\mathbf{a}x - \mathbf{b}\|$ is _____.

Theoretical Solution

We are trying to find a scalar x that minimizes $\|\mathbf{a}x - \mathbf{b}\|$. This is a least-squares problem.

Let us consider,

$$\mathbf{a}x = \mathbf{b} \quad (1)$$

Define the residual vector as,

$$\mathbf{r} = \mathbf{a}x - \mathbf{b} \quad (2)$$

The least-squares problem is to minimize the value:

$$\mathbf{r}^\top \mathbf{r} = (\mathbf{a}x - \mathbf{b})^\top (\mathbf{a}x - \mathbf{b}) \quad (3)$$

$$\mathbf{r}^\top \mathbf{r} = (\mathbf{a}^\top x - \mathbf{b}^\top)(\mathbf{a}x - \mathbf{b}) \quad (4)$$

$$\mathbf{r}^\top \mathbf{r} = \mathbf{a}^\top \mathbf{a}x^2 - \mathbf{a}^\top \mathbf{b}x - \mathbf{b}^\top \mathbf{a}x + \mathbf{b}^\top \mathbf{b} \quad (5)$$

Now, to minimize $\mathbf{r}^\top \mathbf{r}$, define

$$f(x) = \mathbf{a}^\top \mathbf{a}x^2 - 2\mathbf{a}^\top \mathbf{b}x + \mathbf{b}^\top \mathbf{b} \quad (6)$$

Theoretical Solution

Taking gradient with respect to x :

$$\nabla_x f(x) = 2\mathbf{a}^\top \mathbf{a}x - 2\mathbf{a}^\top \mathbf{b} \quad (7)$$

Setting $\nabla_x f(x) = 0$:

$$\mathbf{a}^\top \mathbf{a}x = \mathbf{a}^\top \mathbf{b} \quad (8)$$

$$\implies \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} x = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 3\sqrt{2} \end{pmatrix} \quad (9)$$

$$\implies 2x = 3\sqrt{2} \quad (10)$$

$$\implies x = \frac{3\sqrt{2}}{2} \quad (11)$$

Theoretical Solution

Therefore the vector $\mathbf{r}$ is,

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \frac{3\sqrt{2}}{2} - \begin{pmatrix} 0 \\ 3\sqrt{2} \end{pmatrix} \quad (12)$$

$$\mathbf{r} = \begin{pmatrix} \frac{3\sqrt{2}}{2} \\ -\frac{3\sqrt{2}}{2} \end{pmatrix} \quad (13)$$

$$\|\mathbf{r}\| = \sqrt{\mathbf{r}^T \mathbf{r}} = \sqrt{\begin{pmatrix} \frac{3\sqrt{2}}{2} & -\frac{3\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} \frac{3\sqrt{2}}{2} \\ -\frac{3\sqrt{2}}{2} \end{pmatrix}} = \sqrt{9} \quad (14)$$

$$\|\mathbf{r}\| = 3 \quad (15)$$

Therefore the value $\min_x \|\mathbf{ax} - \mathbf{b}\|$ is 3.

Plot

